Domain and range

1 Consider the function $y = |x|$.

a What values of x can be substituted into the given function?

b What values of y could the function have?

2 Consider the function $y = |x - 3|$.

a What is the least possible value that this function can have?

b What is the greatest possible value that this function can have?

c What is the range of the function?

d What is the domain of the function?

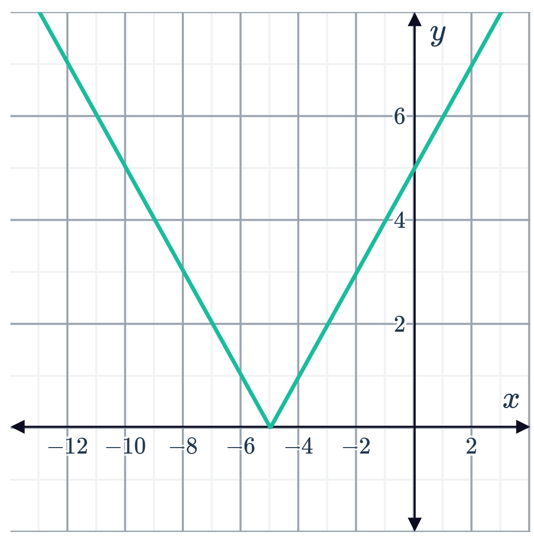
3 Consider the function $y = |x + 5|$.

a What is the least possible value that this function can have?

b What is the greatest possible value that this function can have?

c What is the range of the function?

d What is the domain of the function?

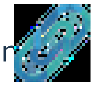


4 The domain of function $f(x)$ is $x \in \mathbb{R}$ and the range of the function is $[2, \infty)$. Write a possible equation of the function.

5 State the range of the following functions in interval notation:

a The function $f(x) = |2x + 2|$ has a domain of $(-2, 2]$.

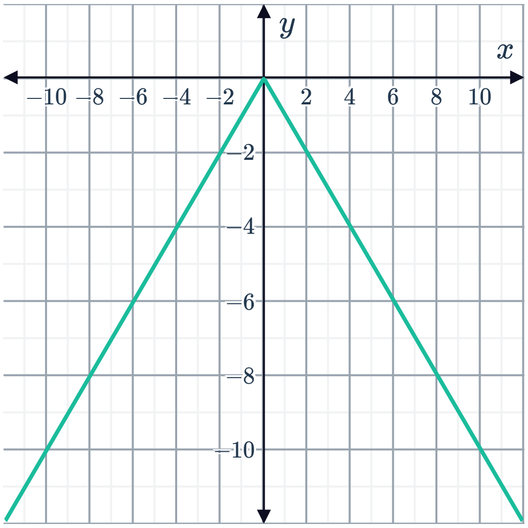
b The function $f(x) = |3x + 6|$ has a domain of $(-1, 0]$.

6 For each of the following graphs of function 

i State the domain of the function.

ii State the range of the function.

a



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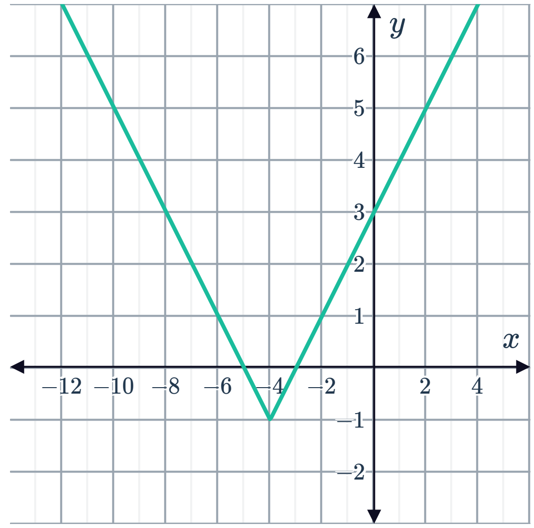

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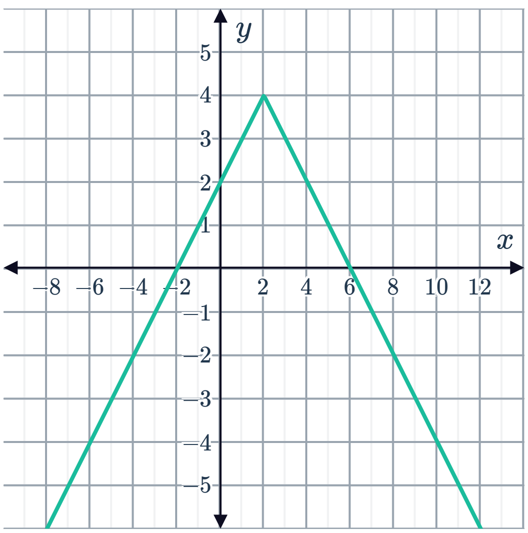

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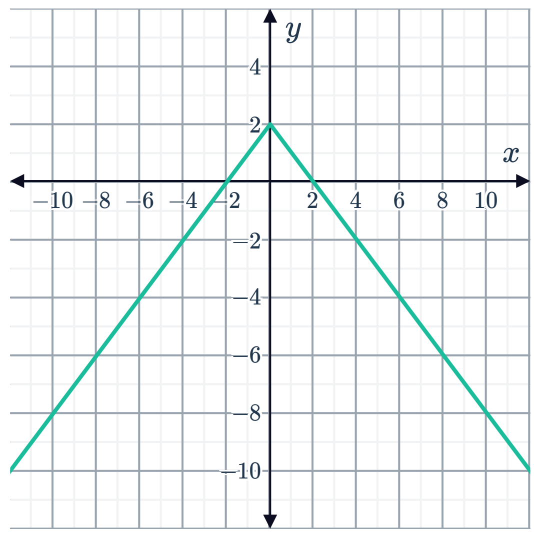
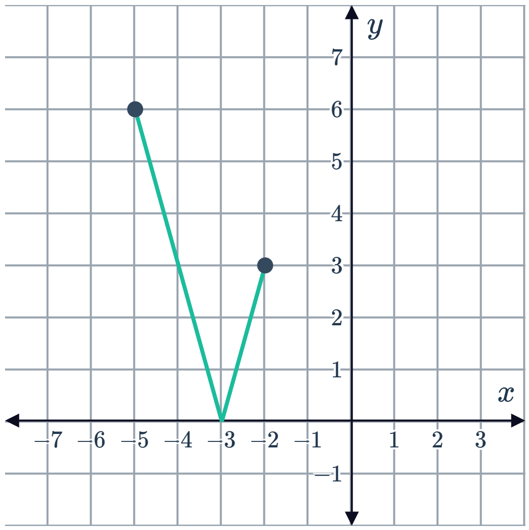


7 For each of the following graph of the funct[redacted](x):

i State the domain of the function.

ii State the range of the function.

a



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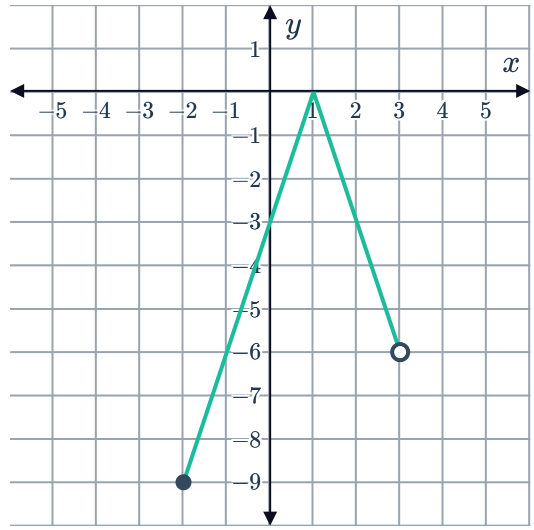

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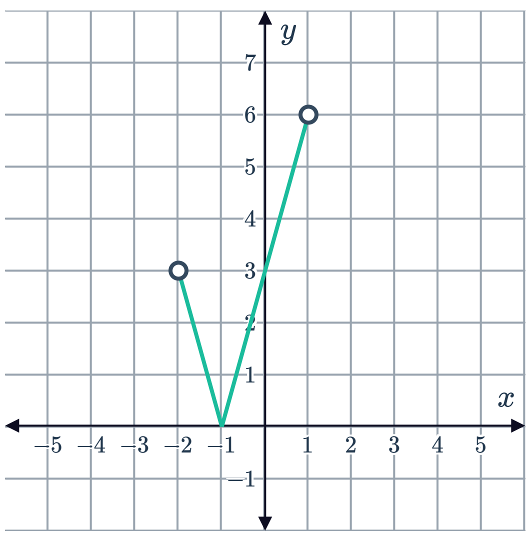

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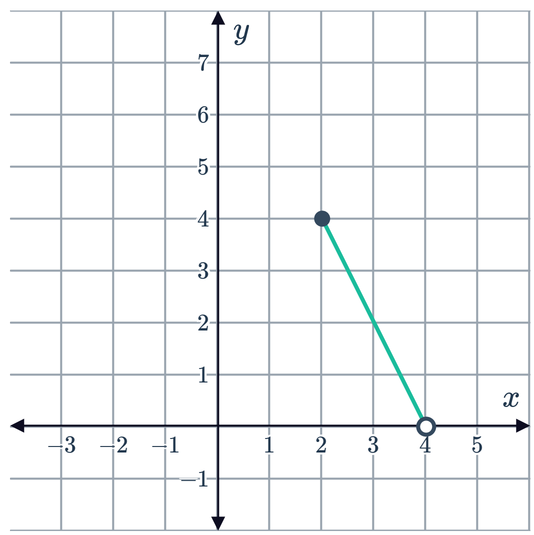


Graphs of absolute value functions

8 For each of the following functions:

i State the domain of the function.

ii State the range of the function.

iii Sketch the graph of the function.

a $y = |x| - 3$

b $y = |x - 2| - 4$

c $y = 4 - 3|x|$

9 For each of the following functions:

i State the the range of the function.

ii Sketch the graph of the function.

iii State the coordinates of the vertex.

iv State the minimum value of the function.

a $y = |x + 3| - 4$

b $y = -|x - 4| + 2$

10 Consider the function $y = |x + 1|$.

a Complete the given table.

b Sketch the graph of the function.

c State the equation of the axis of symmetry.

d State the coordinates of the vertex.

x	-2	-1	0	1	2
y					

e Complete the given table by writing the equation and slope for the two lines that make up the graph of the function.

	Equation	Slope
$x < -1$		
$x > -1$		

11 For each of the following functions:

i Sketch the graph of the function.

ii State the coordinates of the vertex.

iii State the domain.

iv State the range.

a $y = |x|$

b $y = 3|x| - 3$

12 Consider the function $y = 3|x| - 3$.

a Sketch the graph of the function.

b State the equation of the axis of symmetry.

c What are the coordinates of the vertex?

d What are the x-intercepts?

e What is the y-intercept?

f What is the slope of the line to the left of the vertex?

g What is the slope of the line to the right of the vertex?

13 For each of the following functions:

i Does the graph of the function open upwards or downwards?

ii State the coordinates of the vertex.

iii Sketch the graph of the function.

a $y = -\left|x\right|$

b $y = -\left|\frac{x}{7}\right|$

c $y = \left|x - 4\right|$

d $y = \left|2x\right|$

e $y = -\left|x + 5\right|$

f $y = \left|3x - 6\right|$

g $y = \left|2x + 10\right|$

h $y = \left|x + 2\right|$

i $y = \left|-x\right|$

14 Consider the function $f\left(x\right) = \left|x + 1\right|$.

a Does the graph of the function open upwards or downwards?

b State the coordinates of the vertex.

c State the equation of the line of symmetry.

d State whether the following absolute value functions have a narrower graph than $y = \left|x + 1\right|$:

i $y = \left|x + 5\right|$

ii $y = \left|\frac{x}{5} + 1\right|$

iii $y = \left|5x + 1\right|$

15 Consider the function $f\left(x\right) = \left|3 - x\right|$.

a Does the graph of the function open upwards or downwards?

b State the coordinate of the vertex.

c State the equation of the line of symmetry.

d State whether the following absolute value functions have a vertex that is to the left of the vertex of $y = \left|3 - x\right|$:

i $y = \left|x - 3\right|$

ii $y = -\left|3 - x\right|$

iii $y = \left|2 - x\right|$

iv $y = \left|6 - x\right|$

16 Consider the function $y=|6x|$.

a Does the graph of the function open upwards or downwards?

b State the equation of the line of symmetry.

c State the coordinates of the vertex.

d Sketch the graph of the function.

e State whether the following absolute value functions have a narrower graph than $y=|6x|$:

i $y=-|6x|$ ii $y=|4x|$ iii $y=|7x|$ iv $y=|9x|+3$

17 Consider the function $y=|x-4|$.

a Does the graph of the function open upwards or downwards?

b State the equation of the line of symmetry.

c State the coordinates of the vertex.

d Sketch the graph of the function.

e State whether the following absolute value functions have a vertex further to the left than the vertex of $y=|x-4|$:

i $y=|x+4|$ ii $y=|x-5|$

iii $y=|x-4|+3$ iv $y=|x|$

18 An absolute value function $f(x)$ has its vertex at $(4,0)$, and goes through the point $A(9,15)$.

a Find the slope between the vertex and point A.

b State the equation of the function, expressing $f(x)$ in terms of x.

c State the domain of $f(x)$.

d State the range of $f(x)$.

e State whether the following functions have a vertex that is to the left of the vertex of $f(x)=|3(x-4)|$:

i $y=|6x-12|$ ii $y=|6x-24|$ iii $y=|3x-24|$

19 Consider the function $y=3|x-2|$.

a Sketch the graph of the function. b State the coordinates of the vertex.

20 For each of the following functions:

i Sketch the graph of the function.

ii State the coordinates of the vertex.

iii State the minimum value of the function.

a $y = |x+2| + 5$

b $y = -|x-4| + 1$

21 Consider the function $f(x) = 11 - |x|$.

a Sketch the graph of the function.

b What is the maximum value of $f(x)$?

c On which interval is the function increasing?

d On which interval is the function decreasing?

22 Sketch the graph of the following functions:

a $y = |5x|$

b $y = |x-5|$

c $y = -|3x-6|$

d $y = |5-x|$

e $y = \left|\frac{x}{3} - 2\right|$

f $y = 8 - |4x|$

g $y = -3|x|$

h $y = |x| - 2$

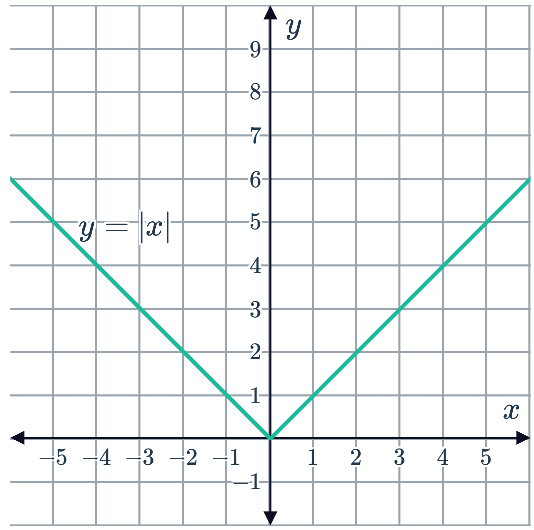
i $y = 2|x+3| - 4$

j $y = \frac{1}{3}|x-4| - 2$

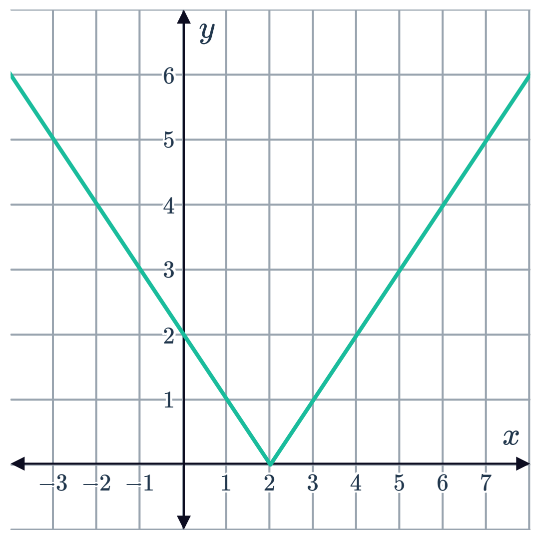
k $y = -\frac{1}{4}|x+2| + 4$

Transformations

23 Use the following graph of $y = |x|$ to graph $y = |x - 4| + 4$.



24 Consider the graph of the absolute value function shown:
If this function is reflected across the y-axis, what is the equation of the new function formed?



25 State the equation of the new graph when the graph of $y = |x|$ undergoes the following transformations:

- **a** Translated 5 units left and 3 units down.
- **b** Stretched vertically by a factor of 4 and reflected about the x-axis.
- **c** Compressed vertically by a factor of $\frac{1}{4}$, reflected about the x-axis and translated 3 units left and 3 units up.

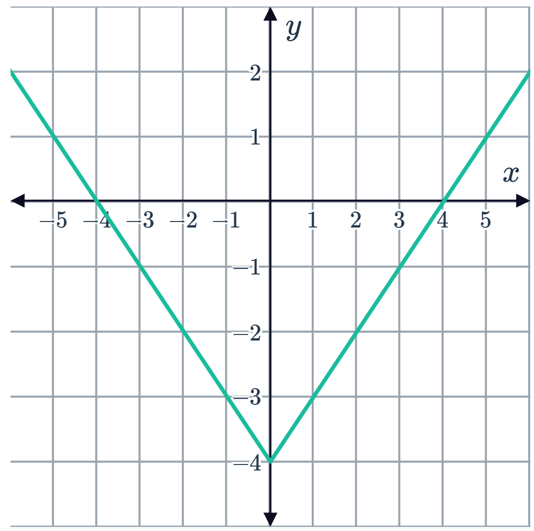
26 Consider the function $f\left(x\right)=|x-2|$.

a If we start with the graph of the function $g\left(x\right)=|x|$, what transformation will produce the graph of $f\left(x\right)$?

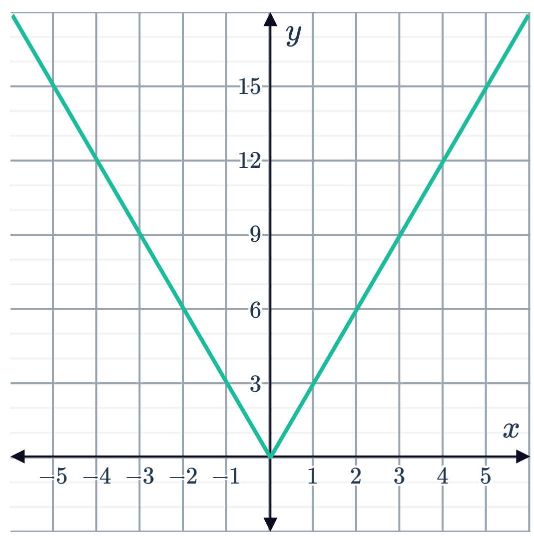
b Are the domains of $f\left(x\right)$ and $g\left(x\right)$ the same?

c Are the ranges of $f\left(x\right)$ and $g\left(x\right)$ the same?

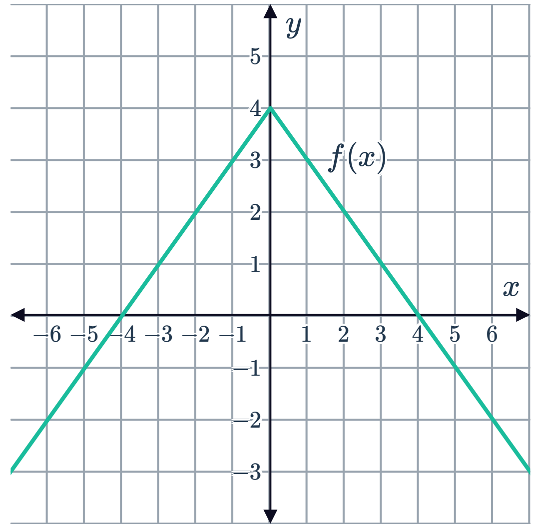
27 Consider the function $y=|x|-4$ that has been graphed. What transformation has been applied to the graph of $y=|x|$ to obtain the graph of $y=|x|-4$?



28 Consider the function $f(x)$ that has been graphed. What transformation has been applied to the graph of $y=|x|$ to obtain the given graph?



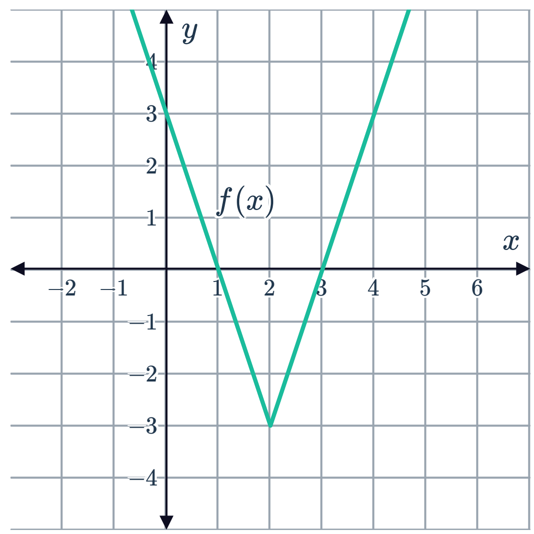
29 Consider the graph of a function $f(x)$:



a What transformation is applied to the graph of $y = |x|$ to obtain the given graph?

b State the equation of $f(x)$ for all real x.

30 Consider the graph of the function $f(x)$:



a State the slope of the function for $x \geq 2$.

b State the slope of the function for $x < 2$.

c State the coordinates of the vertex of the function $y = |x|$.

d State the coordinates of the vertex of the function that has been graphed.

e State the transformation applied to the graph of $y = |x|$ to obtain the given graph.

f State the equation of $f(x)$ for all real x.

31 The table below shows values that satisfy th[redacted]ction $f(x) = |x|$:

x	-3	-2	-1	0	1	2	3
$y = f(x)$	3	2	1	0	1	2	3
$g(x) = 5f(x)$							
$h(x) = -2f(x)$							

a Complete the table of values for each transformation of $f(x)$.

b Sketch the graphs of $f(x)$ and $g(x)$ on the same coordinate plane.

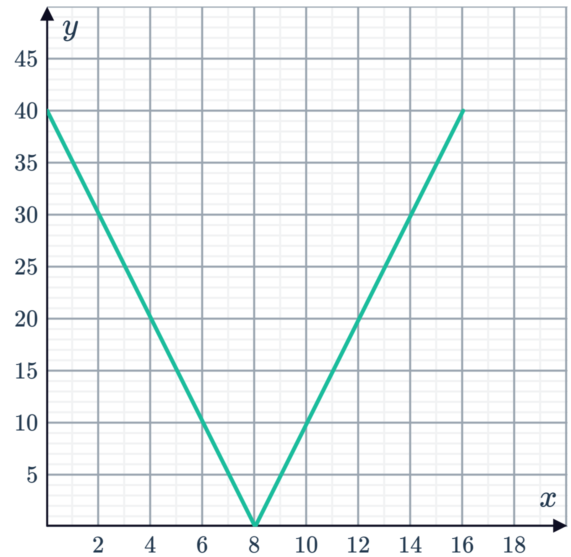
c Describe the transformation undergone by $f(x)$ to become $g(x)$.

d Sketch the graphs of $f(x)$ and $h(x)$ on the same coordinate plane.

e Describe the transformation undergone by $f(x)$ to become $h(x)$.

Applications

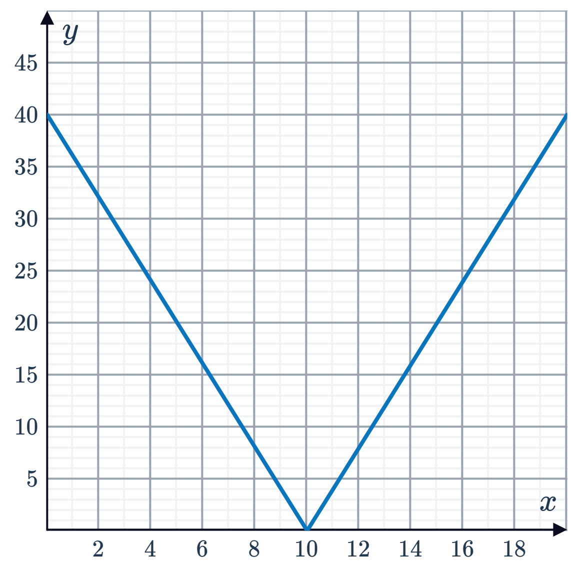
32 Michael is training for the land speed record and takes his new car for a test drive by driving straight down a closed highway and back. His distance y in miles from the end of the highway x minutes after he takes off is given by the function $y = |5x - 40|$ which is graphed below.



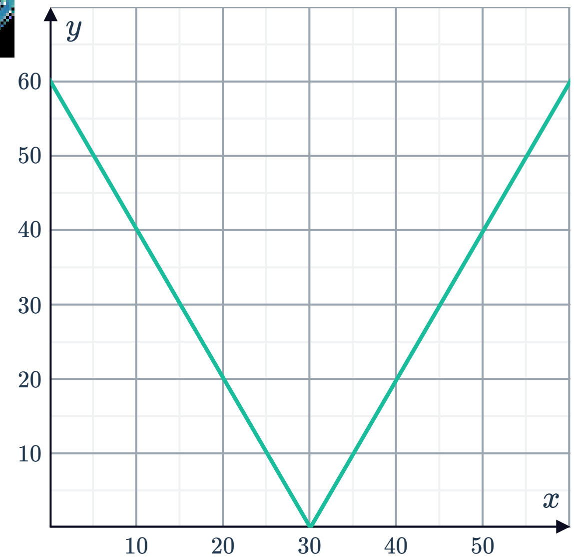
a How far does he drive in total?

b How long does Michael take to reach the end of the highway?

c The next day Michael goes for another drive down the same route and his distance from the end of highway is given by $y = |4x - 40|$ which is graphed. Is Michael driving faster or slower than the previous day?



33 Robert is taking his new car for a test drive by driving straight down the highway and back. His distance y in km from the end of a 60 km highway x minutes after he takes off is given by the function $y = |2x - 60|$ shown in the first graph:

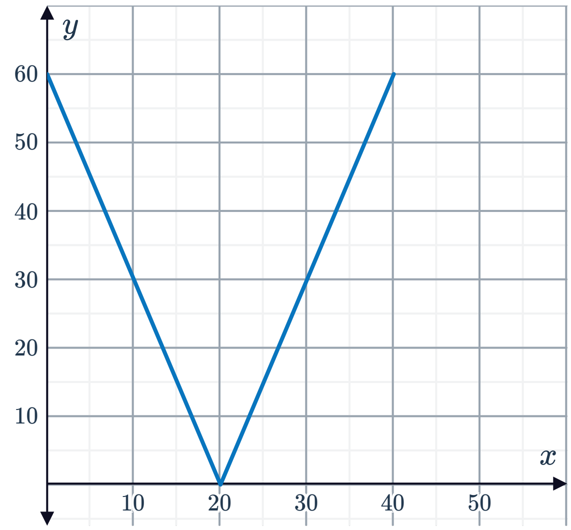


a In this context, what is the domain of the function?

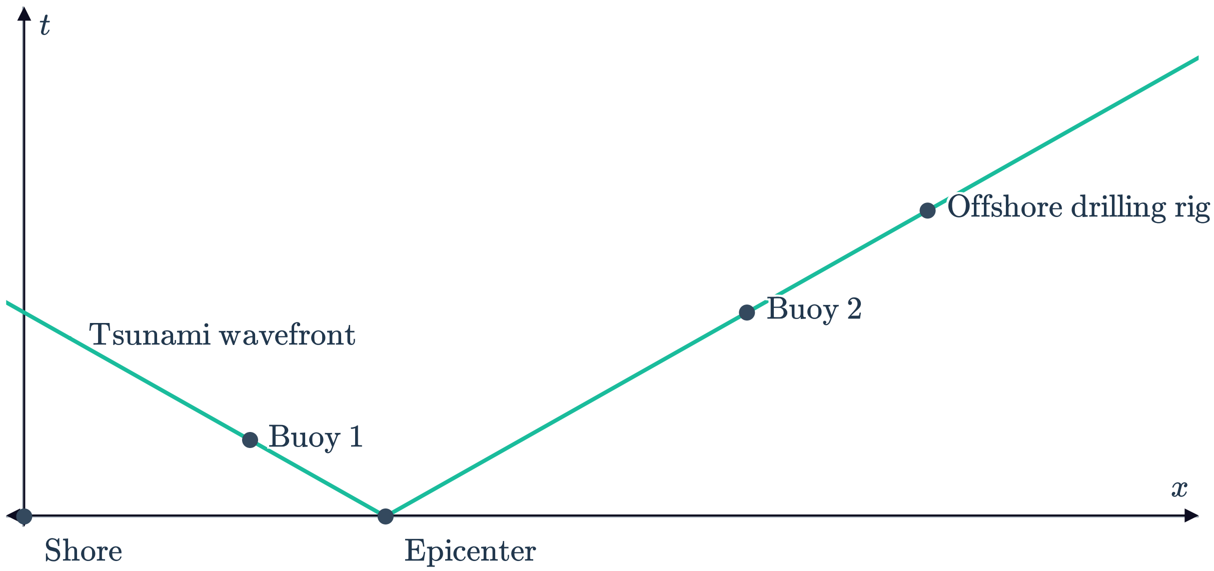
b How far does he drive in total?

c How long does Robert take to reach the end of the highway?

d The next day Robert goes for another drive down the same route and his distance from the end of highway is given by $y = |3x - 60|$ which is shown in the second graph:
Is Robert driving faster or slower than the previous day? Explain your answer.



34 A deep sea landslide is detected at its epice[redacted] 1050 mi offshore, at time $t = 0$ minutes. Thirty minutes later, the tsunami is detected at a wave buoy 1 which is 700 mi from shore. After 90 minutes the tsunami is detected at buoy 2 which is 2100 mi from shore.



a How fast is the wavefront of the tsunami moving?

b State the equation of the wavefront of the tsunami in terms of the distance x mi from shore and the time t minutes after the underwater landslide.

c Measuring from the initial detection at the epicenter ($t = 0$), how long will it take the tsunami to reach the shore in minutes?

d How many hours will it take the tsunami to reach the offshore drilling rig if it is located 2800 mi from the shore?

35 In a game show 'The Cost is Lost', contestants need to guess the price of several items. Each contestant starts with 300 points and each time they guess the price of an item, they lose 3 times the difference between their guess and the actual price.

a The first item is a guitar tuner which costs $\$13$.
If x is the price in dollars guessed for this item, write an equation for y, the points lost for that guess.

b Graph the equation you found in part (a).

c The second item is a bluetooth speaker which costs $\$31$.
If x is the price in dollars guessed for this item, write an equation for y, the points lost for that guess.

d Graph the equation you found in part (c) on the same number plane as part (b).

e State whether the following guesses for the price of the guitar tuner would make you lose more points than if you guessed the same amount for the price of the bluetooth speaker.

i $\$33$ **ii** $\$5$ **iii** $\$24$ **iv** $\$18$